By submitting this assignment, I agree to the following:
"Aggies do not lie, cheat, or steal, or tolerate those who do."
"I have not given or received any unauthorized aid on this assignment."
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Section: 571
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Hierarchy

Program

- Define dictionary with ASCII characters
    - Numbers, letters, and symbols to ASCII
- User input
    - Read input from user
        - Read input for the time in 24-hour format
        - Read input for the clock type
        - Read input for the preferred character
    - Handle user input
        - Convert 24 hour time to 12 hour if the user chooses the 12 hour format
        - Determine whether it is AM or PM
            - Concatenate AM or PM to the time
        - If the preferred character is either of the following characters: abcdeghkmnopqrsuvwxyz@$&*=, it is permitted
        - If the user does not input a preferred character, use the digits as the characters.
        - If the time will have a leading zero or not
- Display time in ASCII art
    - Convert each character to ASCII
        - Numbers to ASCII based on preferred character
        - Letters and symbols to ASCII from dictionary
    - Store each ASCII character in a list
    - Combine ASCII characters into a single string
    - Output the string

Variable List

- character_to_ascii
    - Dictionary
    - Keys are characters
    - Values are list with characters per line
- time
    - String
    - The time in 24-hour format that will be displayed

- clock_type
    - Integer that either is 12 or 24
    - If 12, time is displayed in 12-hour format
    - If 24, time is displayed in 24-hour format
- preferred_character
    - String
    - Character that is in “abcdeghkmnopqrsuvwxyz@$&*=”
- ascii_characters
    - List containing all the characters converted to ASCII art in order of display
- ascii_string
    - String that will be printed and display the ASCII time

Test Cases

Inputs	Outputs
Time: 3:33 Clock type: 24 Preferred character:	333 333 333 3 : 3 3 333 333 333 3 : 3 3 333 333 333
Time: 3:33 Clock type: 12 Preferred character:	333 333 333 A M M 3 : 3 3 A A MM MM 333 333 333 AAA M M M 3 : 3 3 A A M M 333 333 333 A A M M
Time: 13:00 Clock type: 12 Preferred character: *	* *** *** PPP M M ** : * * * * P P MM MM * * * * * PPP M M M * : * * * * P M M *** *** *** P M M
Time: 13:00 Clock type: 24 Preferred character: *	* *** *** *** ** * : * * * * * *** * * * * * * : * * * * *** *** *** ***
Time: 18:42 Clock Type: 12 Preferred Character:	666 4 4 222 PPP M M 6 : 4 4 2 P P MM MM 666 444 222 PPP M M M 6 6 : 4 2 P M M 666 4 222 P M M

Time: 18:42 Clock Type: 13 Preferred Character:	`Clock type must be 12 or 24. Try again:`
Time: ASDF Clock Type: 12 Preferred Character:	`Not a valid time! Try again:`
Time: 12:00 Clock Type: 12 Preferred Character: (	`Character not permitted! Try again:`
Time: 24:00 Clock Type: 24 Preferred Character:	`Not a valid time! Try again:`
Time: 12 Clock Type: 24 Preferred Character:	`Not a valid time! Try again:`

Part G

Describe the difficulty your team had when combining the code at the end. How could your team have specified the design more clearly at the beginning?

- One difficulty we had when combining the code is that we had different ways of storing data. For example, to store each of the ascii characters, Leo used a string with newline escape characters, while David used a list with each element representing a row. Another difficulty is that we had different variable names. For example, for the dictionary mapping characters to ascii, Leo called it ascii, while David called character_to_ascii. We could have specified the design more clearly in the beginning by breaking it down into smaller parts, focusing more on how we would store the data, and listing all the variables that we would use.

Describe any benefits and drawbacks you saw by dividing the coding like this. Can you see reasons why this might be a good idea? How about a bad idea?

- A benefit of dividing the code like this is we can complete a program more quickly because we are working on different parts at the same time. Another advantage is that we can make sure specific parts are working correctly before it is combined. One drawback is that our parts might have a different structure, organization, or variable names, which can make it harder to combine into one. Another drawback is that we have to wait for the other parts to be finished before we can combine.